Table S10 Directional discrimination and paired scoring-workflow comparisons

A. Method-specific estimates

Target pair	Scheme	n (dual/A/B)	AUROC B [95% CI]	AUROC A [95% CI]	Min. AUROC [95% CI]
EGFR/HER2	S1	28 / 37 / 32	0.6274 [0.4865, 0.7645]	0.2467 [0.1328, 0.3783]	0.2467 [0.1328, 0.3783]
EGFR/HER2	S2	28 / 37 / 32	0.3731 [0.2355, 0.5150]	0.2210 [0.0993, 0.3538]	0.2210 [0.0993, 0.3516]
EGFR/HER2	S3	28 / 37 / 32	0.5241 [0.3793, 0.6641]	0.2634 [0.1339, 0.4051]	0.2634 [0.1339, 0.4051]
EGFR/HER2	S4	28 / 37 / 32	0.4204 [0.2760, 0.5690]	0.2812 [0.1551, 0.4208]	0.2812 [0.1551, 0.4163]
EGFR/HER2	S5	28 / 37 / 32	0.4344 [0.2896, 0.5820]	0.2455 [0.1239, 0.3806]	0.2455 [0.1239, 0.3806]
JAK1/JAK2	S1	32 / 32 / 32	0.5908 [0.4512, 0.7275]	0.7485 [0.6235, 0.8604]	0.5908 [0.4512, 0.7246]
JAK1/JAK2	S2	32 / 32 / 32	0.4385 [0.2979, 0.5791]	0.7402 [0.6055, 0.8623]	0.4385 [0.2979, 0.5791]
JAK1/JAK2	S3	32 / 32 / 32	0.4736 [0.3291, 0.6211]	0.7480 [0.6162, 0.8623]	0.4736 [0.3291, 0.6201]
JAK1/JAK2	S4	32 / 32 / 32	0.4443 [0.2988, 0.5898]	0.6846 [0.5430, 0.8184]	0.4443 [0.2988, 0.5898]
JAK1/JAK2	S5	32 / 32 / 32	0.5137 [0.3682, 0.6592]	0.7012 [0.5605, 0.8311]	0.5137 [0.3682, 0.6592]
JAK1/TYK2	S1	32 / 32 / 32	0.6475 [0.5049, 0.7812]	0.4219 [0.2812, 0.5654]	0.4219 [0.2812, 0.5635]
JAK1/TYK2	S2	32 / 32 / 32	0.6631 [0.5225, 0.7940]	0.5596 [0.4102, 0.7021]	0.5596 [0.4102, 0.6836]
JAK1/TYK2	S3	32 / 32 / 32	0.7090 [0.5742, 0.8271]	0.4873 [0.3379, 0.6338]	0.4873 [0.3379, 0.6309]
JAK1/TYK2	S4	32 / 32 / 32	0.7637 [0.6348, 0.8779]	0.6758 [0.5254, 0.8164]	0.6758 [0.5254, 0.7998]
JAK1/TYK2	S5	32 / 32 / 32	0.6689 [0.5322, 0.8067]	0.6846 [0.5430, 0.8115]	0.6689 [0.5176, 0.7666]
PIK3CA/mTOR	S1	32 / 32 / 32	0.7363 [0.6074, 0.8486]	0.5801 [0.4346, 0.7188]	0.5801 [0.4346, 0.7129]
PIK3CA/mTOR	S2	32 / 32 / 32	0.5752 [0.4297, 0.7168]	0.6592 [0.5195, 0.7871]	0.5752 [0.4287, 0.6982]
PIK3CA/mTOR	S3	32 / 32 / 32	0.7041 [0.5674, 0.8281]	0.7305 [0.5986, 0.8467]	0.7041 [0.5596, 0.7998]
PIK3CA/mTOR	S4	32 / 32 / 32	0.5166 [0.3730, 0.6602]	0.6641 [0.5263, 0.7910]	0.5166 [0.3730, 0.6523]
PIK3CA/mTOR	S5	32 / 32 / 32	0.5977 [0.4492, 0.7344]	0.7148 [0.5801, 0.8350]	0.5977 [0.4482, 0.7285]
AChE/BChE	S1	27 / 25 / 28	0.6341 [0.4756, 0.7867]	0.6442 [0.4907, 0.7871]	0.6341 [0.4607, 0.7540]
AChE/BChE	S2	27 / 25 / 27	0.5052 [0.3392, 0.6667]	0.4925 [0.3333, 0.6502]	0.4925 [0.3141, 0.5867]
AChE/BChE	S3	27 / 25 / 28	0.6474 [0.4903, 0.7881]	0.4907 [0.3360, 0.6468]	0.4907 [0.3360, 0.6296]
AChE/BChE	S4	27 / 25 / 27	0.5437 [0.3822, 0.6978]	0.3909 [0.2401, 0.5528]	0.3909 [0.2401, 0.5336]
AChE/BChE	S5	27 / 27 / 28	0.6228 [0.4650, 0.7668]	0.4511 [0.2923, 0.6111]	0.4511 [0.2923, 0.5967]
F2/F10	S1	32 / 32 / 32	0.4082 [0.2656, 0.5557]	0.3623 [0.2363, 0.5049]	0.3623 [0.2305, 0.4805]
F2/F10	S2	31 / 32 / 32	0.5383 [0.3911, 0.6774]	0.5675 [0.4194, 0.7067]	0.5383 [0.3821, 0.6300]
F2/F10	S3	32 / 32 / 32	0.6094 [0.4668, 0.7490]	0.6230 [0.4785, 0.7568]	0.6094 [0.4531, 0.7021]
F2/F10	S4	31 / 32 / 32	0.6341 [0.4919, 0.7641]	0.5696 [0.4173, 0.7127]	0.5696 [0.4163, 0.6855]
F2/F10	S5	31 / 32 / 32	0.5625 [0.4133, 0.7107]	0.5554 [0.4062, 0.7026]	0.5554 [0.3921, 0.6633]

AUROC, target B: dual versus A-only; AUROC, target A: dual versus B-only. Counts: dual/A-only/B-only. S1, Vina; S2, fixed-pose GNINA; S3, fixed-pose RTMScore; S4, maximum saved-pose CNNscore; S5, independent GNINA. Higher scores are favorable. Paired contrasts use their own shared members and need not equal differences between estimates in panel A. CI: 95% class-stratified percentile bootstrap (10,000 resamples; seed 271828). S4-S2: saved-pose reranking. Supporting S4-S1 in Table S9 has no added CI.

Table S10 Directional discrimination and paired scoring-workflow comparisons (continued)

A. Method-specific estimates

Target pair	Scheme	n (dual/A/B)	AUROC B [95% CI]	AUROC A [95% CI]	Min. AUROC [95% CI]
PPARG/PPARA	S1	32 / 32 / 32	0.6553 [0.5117, 0.7861]	0.6021 [0.4614, 0.7402]	0.6021 [0.4561, 0.7051]
PPARG/PPARA	S2	30 / 32 / 32	0.6583 [0.5177, 0.7885]	0.2802 [0.1573, 0.4156]	0.2802 [0.1573, 0.4156]
PPARG/PPARA	S3	32 / 32 / 32	0.6455 [0.5029, 0.7744]	0.3105 [0.1748, 0.4492]	0.3105 [0.1748, 0.4492]
PPARG/PPARA	S4	30 / 32 / 32	0.5781 [0.4313, 0.7229]	0.3312 [0.2000, 0.4719]	0.3312 [0.2000, 0.4698]
PPARG/PPARA	S5	32 / 32 / 32	0.5146 [0.3701, 0.6543]	0.4463 [0.3066, 0.5957]	0.4463 [0.3037, 0.5596]
PPARA/PPARD	S1	32 / 32 / 32	0.6787 [0.5430, 0.8047]	0.5107 [0.3652, 0.6543]	0.5107 [0.3652, 0.6475]
PPARA/PPARD	S2	27 / 32 / 32	0.7002 [0.5590, 0.8287]	0.6817 [0.5417, 0.8113]	0.6817 [0.5255, 0.7546]
PPARA/PPARD	S3	32 / 32 / 32	0.7520 [0.6240, 0.8653]	0.6689 [0.5332, 0.7979]	0.6689 [0.5322, 0.7725]
PPARA/PPARD	S4	27 / 32 / 32	0.7581 [0.6296, 0.8738]	0.5625 [0.4144, 0.7118]	0.5625 [0.4144, 0.7025]
PPARA/PPARD	S5	32 / 32 / 32	0.6338 [0.4922, 0.7676]	0.6670 [0.5264, 0.7998]	0.6338 [0.4844, 0.7197]

AUROC, target B: dual versus A-only; AUROC, target A: dual versus B-only. Counts: dual/A-only/B-only. S1, Vina; S2, fixed-pose GNINA; S3, fixed-pose RTMScore; S4, maximum saved-pose CNNscore; S5, independent GNINA. Higher scores are favorable. Paired contrasts use their own shared members and need not equal differences between estimates in panel A. CI: 95% class-stratified percentile bootstrap (10,000 resamples; seed 271828). S4-S2: saved-pose reranking. Supporting S4-S1 in Table S9 has no added CI.

Table S10 Directional discrimination and paired scoring-workflow comparisons

B. Paired differences in minimum AUROC

Target pair	Contrast	n (dual/A/B)	Comparison minimum AUROC	Reference minimum AUROC	Difference (95% CI)
EGFR/HER2	S2-S1	28 / 37 / 32	0.2210	0.2467	-0.0257 [-0.1775, 0.1283]
EGFR/HER2	S3-S1	28 / 37 / 32	0.2634	0.2467	0.0167 [-0.1340, 0.1685]
EGFR/HER2	S4-S2	28 / 37 / 32	0.2812	0.2210	0.0603 [-0.0536, 0.1797]
EGFR/HER2	S5-S1	28 / 37 / 32	0.2455	0.2467	-0.0011 [-0.1629, 0.1596]
JAK1/JAK2	S2-S1	32 / 32 / 32	0.4385	0.5908	-0.1523 [-0.3672, 0.0605]
JAK1/JAK2	S3-S1	32 / 32 / 32	0.4736	0.5908	-0.1172 [-0.2861, 0.0498]
JAK1/JAK2	S4-S2	32 / 32 / 32	0.4443	0.4385	0.0059 [-0.1328, 0.1406]
JAK1/JAK2	S5-S1	32 / 32 / 32	0.5137	0.5908	-0.0771 [-0.2920, 0.1396]
JAK1/TYK2	S2-S1	32 / 32 / 32	0.5596	0.4219	0.1377 [-0.0371, 0.2969]
JAK1/TYK2	S3-S1	32 / 32 / 32	0.4873	0.4219	0.0654 [-0.0791, 0.2109]
JAK1/TYK2	S4-S2	32 / 32 / 32	0.6758	0.5596	0.1162 [-0.0127, 0.2539]
JAK1/TYK2	S5-S1	32 / 32 / 32	0.6689	0.4219	0.2471 [0.0312, 0.4082]
PIK3CA/mTOR	S2-S1	32 / 32 / 32	0.5752	0.5801	-0.0049 [-0.1924, 0.1680]
PIK3CA/mTOR	S3-S1	32 / 32 / 32	0.7041	0.5801	0.1240 [-0.0469, 0.2607]
PIK3CA/mTOR	S4-S2	32 / 32 / 32	0.5166	0.5752	-0.0586 [-0.1660, 0.0547]
PIK3CA/mTOR	S5-S1	32 / 32 / 32	0.5977	0.5801	0.0176 [-0.1807, 0.2178]
AChE/BChE	S2-S1	27 / 25 / 27	0.4925	0.6341	-0.1416 [-0.3676, 0.0489]
AChE/BChE	S3-S1	27 / 25 / 28	0.4907	0.6341	-0.1433 [-0.2608, 0.0172]
AChE/BChE	S4-S2	27 / 25 / 27	0.3909	0.4925	-0.1015 [-0.2215, 0.0923]
AChE/BChE	S5-S1	27 / 25 / 28	0.4511	0.6341	-0.1830 [-0.4026, 0.0715]
F2/F10	S2-S1	31 / 32 / 32	0.5383	0.3740	0.1643 [-0.0141, 0.3054]
F2/F10	S3-S1	32 / 32 / 32	0.6094	0.3623	0.2471 [0.0654, 0.3896]
F2/F10	S4-S2	31 / 32 / 32	0.5696	0.5383	0.0312 [-0.0796, 0.1724]
F2/F10	S5-S1	31 / 32 / 32	0.5554	0.3740	0.1815 [-0.0071, 0.3357]
PPARG/PPARA	S2-S1	30 / 32 / 32	0.2802	0.5891	-0.3089 [-0.4688, -0.1313]
PPARG/PPARA	S3-S1	32 / 32 / 32	0.3105	0.6021	-0.2915 [-0.4380, -0.1104]
PPARG/PPARA	S4-S2	30 / 32 / 32	0.3312	0.2802	0.0510 [-0.0719, 0.1740]
PPARG/PPARA	S5-S1	32 / 32 / 32	0.4463	0.6021	-0.1558 [-0.3457, 0.0459]
PPARA/PPARD	S2-S1	27 / 32 / 32	0.6817	0.5625	0.1192 [-0.0706, 0.2697]
PPARA/PPARD	S3-S1	32 / 32 / 32	0.6689	0.5107	0.1582 [0.0059, 0.2979]
PPARA/PPARD	S4-S2	27 / 32 / 32	0.5625	0.6817	-0.1192 [-0.2616, 0.0868]
PPARA/PPARD	S5-S1	32 / 32 / 32	0.6338	0.5107	0.1230 [-0.0762, 0.2754]

AUROC, target B: dual versus A-only; AUROC, target A: dual versus B-only. Counts: dual/A-only/B-only. S1, Vina; S2, fixed-pose GNINA; S3, fixed-pose RTMScore; S4, maximum saved-pose CNNscore; S5, independent GNINA. Higher scores are favorable. Paired contrasts use their own shared members and need not equal differences between estimates in panel A. CI: 95% class-stratified percentile bootstrap (10,000 resamples; seed 271828). S4-S2: saved-pose reranking. Supporting S4-S1 in Table S9 has no added CI.